



Biochemistry of Bones Metabolism

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LEARNING OBJECTIVES

Biochemistry of calcium & Bones

- Chemical composition of bone
- Role of osteoblasts and osteoclasts cells in bone mineralization and bone resorption
- Role of hormone ; PTH, vitamin D and Calcitonin in bone metabolism
- Rickets, Osteomalacia, Osteoporosis, Osteogenesis imperfecta

Biochemistry of Bones

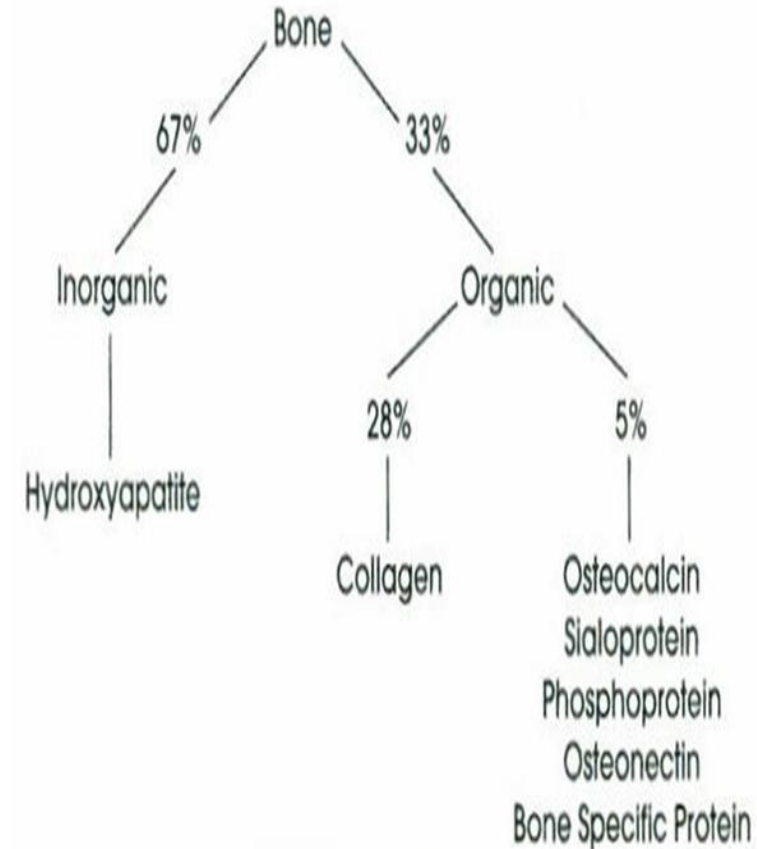
Bone is a dynamic tissue
compose of both organic and
inorganic material

1) organic matrix (33%)

w/c predominantly protein i.e.

- i. Type – I Collagen fibers → 28%
- ii. small amounts of non-proteins
collagenous such as
 - ✓ **Ground Substance: ECF , osteocalcin (OC), glycoproteins and proteoglycans → 5%**

2) inorganic components (67%)

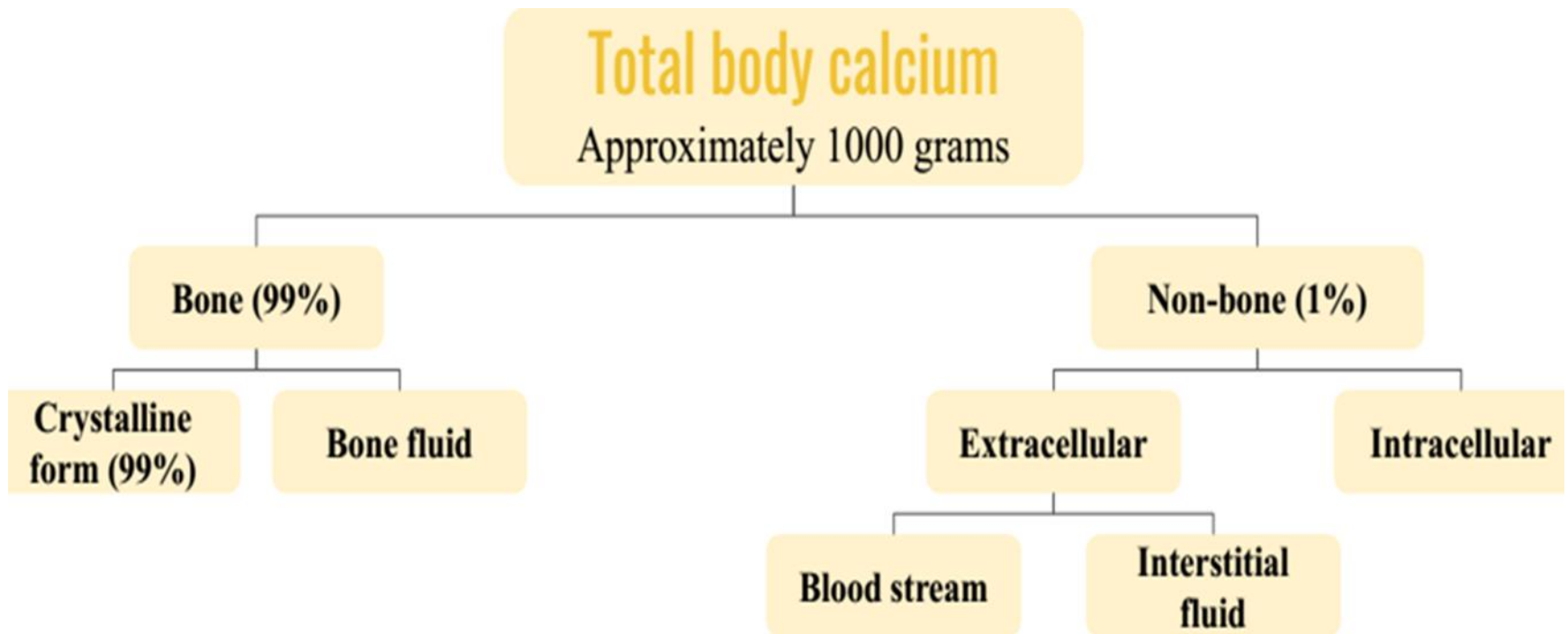


Biochemistry of Bones

- The inorganic component mainly calcium and phosphate w/c form crystalline
 - hydroxyapatite $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$
- Inorganic crystals which mineralize the organic matrix

Biochemistry of Bones...

- Approximately 99% of body calcium is contained in bones
- Bone is dynamic structure that under goes continuing cycles of remodeling, consisting of
 - resorption followed by deposition of new bone tissues

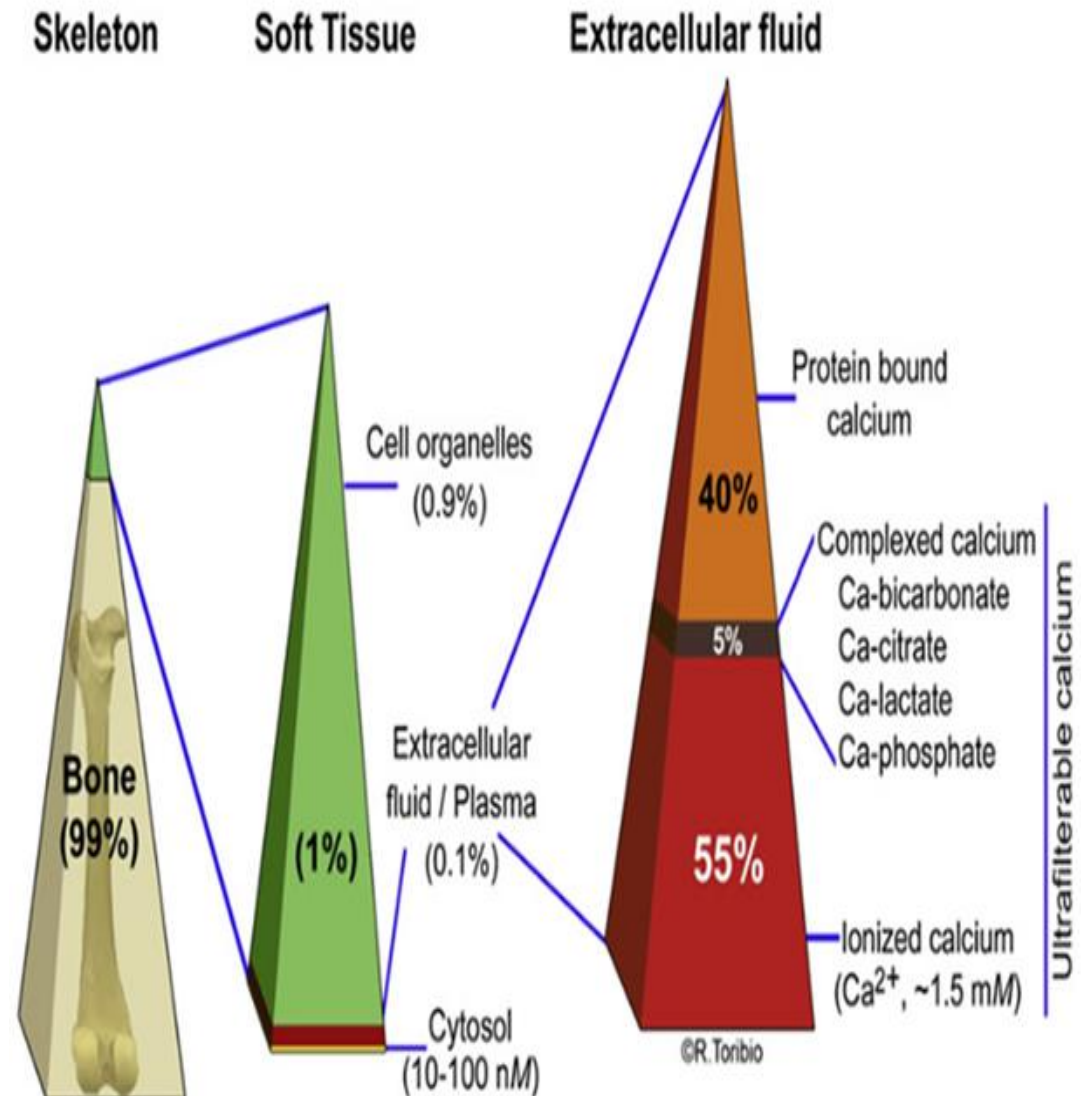


Calcium distribution in the body

- About 99% of total body Ca^{+2} is in the bone (skeleton) and teeth,
- with the remaining 1% present resides in
 - the extracellular fluid (0.1%) and
 - cellular organelles (0.9%)
- **Intracellular Ca^{+2}** concentration is very low , yet it is essential for several critical functions such as:
 - signal transduction , -Nerve conduction
 - muscle contraction, and blood coagulation

Forms of Circulating Ca^{2+} exists

- Ca^{2+} in the skeleton is complexed with phosphorus mainly as hydroxyapatite, which gives the bone its mechanical characteristics



Forms of Circulating Ca^{2+} exists

Calcium distribution in the body

- The three forms of Circulating Ca^{2+} exists in **extracellular fluid** :
 - ionized (free; 55%)
 - protein-bound (mostly to albumin and less to globulins; 40%), and
 - complexed (bound to oxalate, citrate, carbonate, and phosphate; 5%)
- Both ionized and complexed Ca^{+2} are diffusible , while protein-bound Ca^{+2} is not

Bone remodeling

- Bone remodeling is a cyclical process essential for skeletal growth & comprises four phases:
 - ❖ Resorption, reversal (deposition of new bone tissues), bone formation and a resting phase

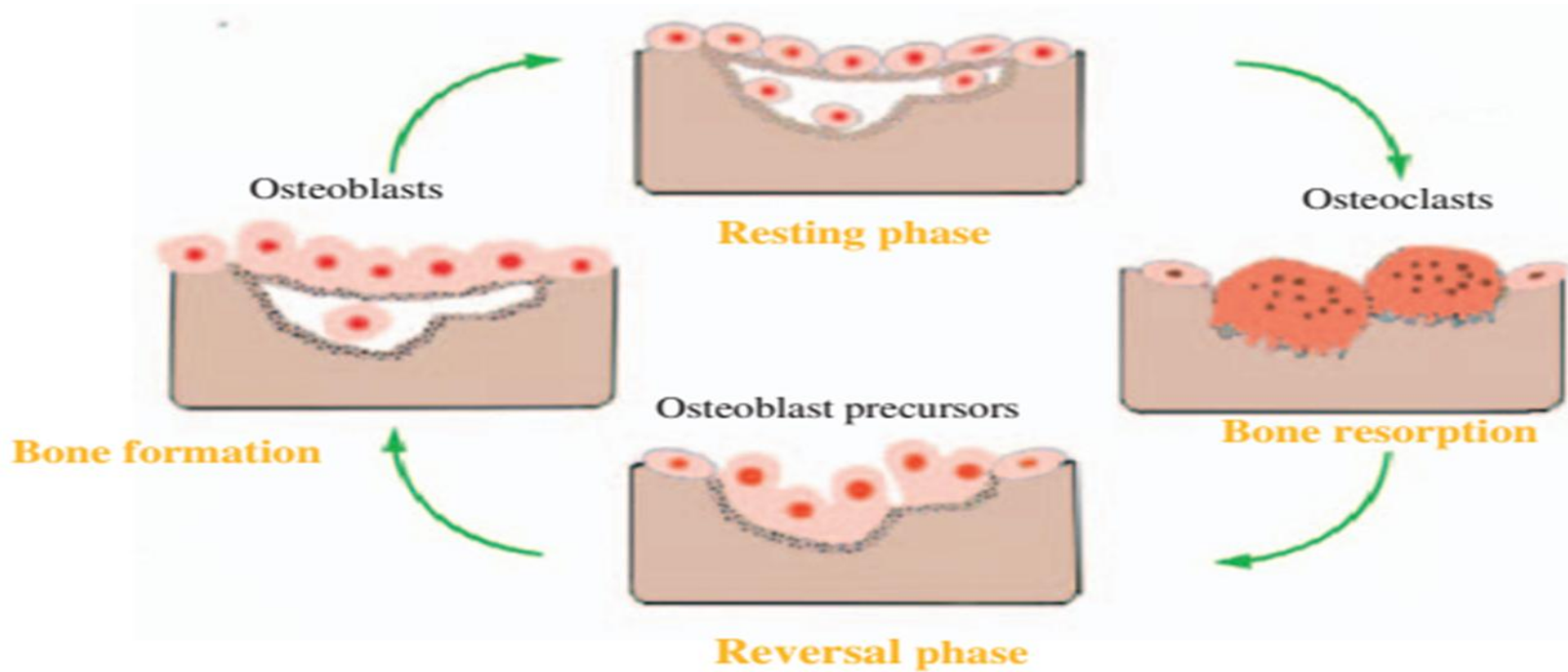


Figure 1. Stages of bone remodelling. Osteoblasts mineralize new bone matrix, while osteoclasts are responsible for bone resorption.

Bone remodeling...

- In a typical cycle, the resorption phase lasts about 7–10 days while bone formation takes up to 2–3 months
- The rate of bone remodelling/cycle is influenced by
 - **mechanical stress, systemic hormones**

Bone remodeling...

Function of Osteoclasts

- Dissolution of both **mineral or calcium salts and enzymatic breakdown** of the organic matrix of bone
- It is multinucleated cells, derived from precursor cells of the monocyte lineage
- These cells express a number of lysosomal enzymes, known as Lysosomal acid proteases w/c
 - ✓ work under acidic environment (PH less 4.0), around the resorption cavities
 - w/c increasing the solubility of hydroxyapatite & allowing demineralization to occur
 - ✓ Excessive dissolution of the mineral phase of bone can lead to hypercalcaemia.

OSTEOBLASTS

- Osteoblasts are mono-nuclear cells;
- It synthesize most of proteins found in bone as well as various growth factors & cytokines
- They are responsible for the deposition of new bone matrix (Osteoid) and its subsequent mineralization
- Osteoblasts control mineralization by
 - regulating the passage of calcium & phosphate ions across their surface membranes
- The Latter contains alkaline phosphatase, which is used to generate phosphate ions from inorganic phosphate

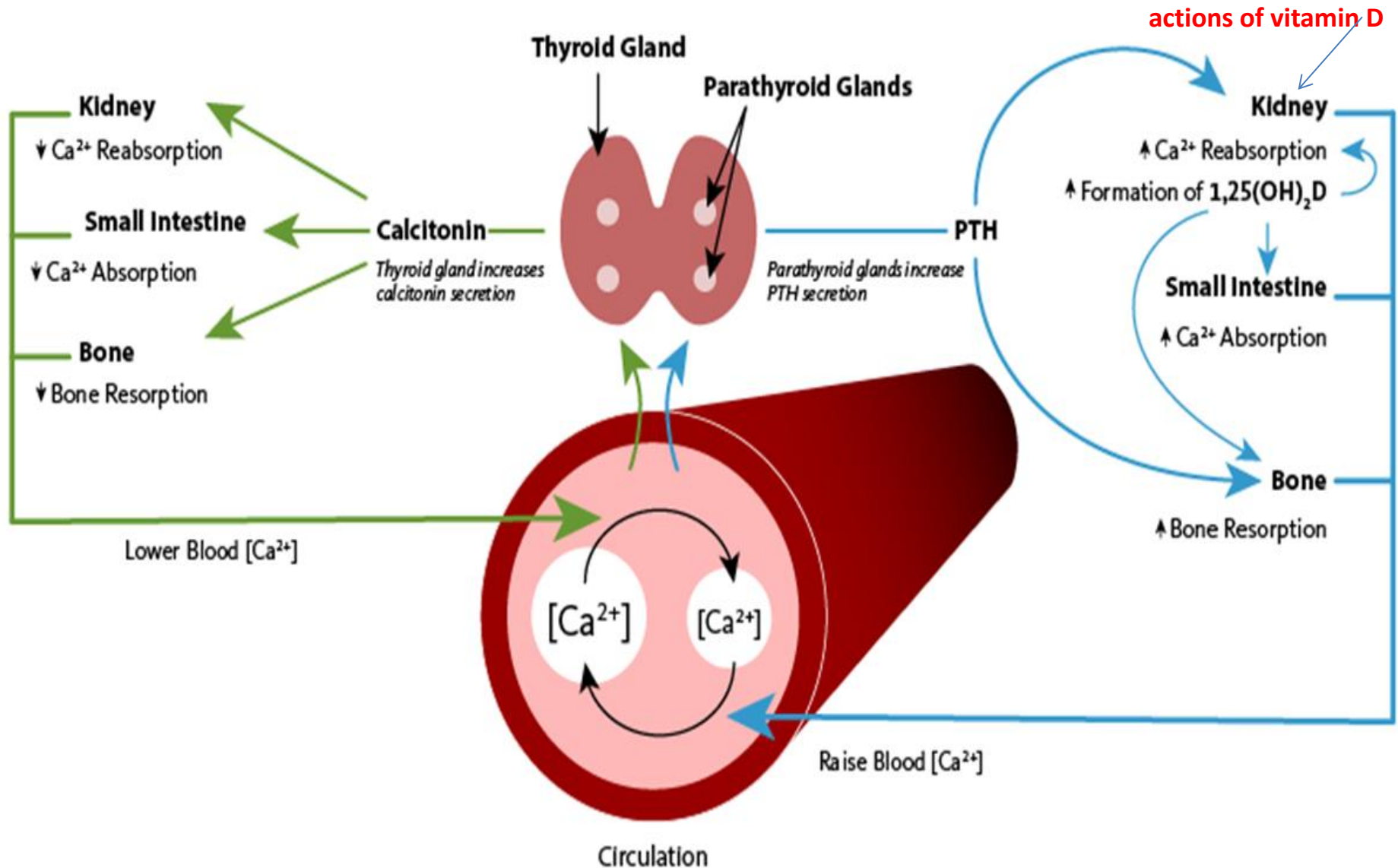
the regulation of bone metabolism

Factors involved in the regulation

- It including the critical organs involved (bone , parathyroids, GI tract, kidneys etc.) as well as
- the hormones (PTH, vitamin D, calcitonin) that regulate calcium and phosphate
 - Parathyroid hormone & prolactin also stimulate Osteoclasts,
 - where as calcitonin & estrogen inhibit them

Calcium and phosphate homeostasis

Figure 1. Calcium Homeostasis



Calcium and phosphate homeostasis ...

- Calcium and phosphorus is regulation of by the actions of
 - many ions and hormones on the intestine, kidneys and bone.
- Calcium and phosphate in the body react in opposite ways:
 - Mostly as blood serum calcium levels rise, phosphate levels fall
- when the plasma Ca^{2+} level is high, PTH secretion is inhibited and the Ca^{2+} is deposited in bones
- when Ca^{2+} is low, secretion PTH is increased and Ca^{2+} is mobilized from the bones

Calcium and phosphate homeostasis ...

- You need the right amount of both calcium and phosphorus for bone health
- Phosphorus also plays an important structural role in
 - nucleic acids and
 - cell membranes
 - it's involved in the body's energy production

Calcium and phosphorus ...

- Human PTH is a linear polypeptide that contains 84 a.a residues
- PTH act directly on bones to increase bone resorption & mobilize Ca^{2+}
- It increasing plasma Ca^{2+} and decreasing the plasma phosphate,
 - PTH increases phosphate excretion in urine
- This **phosphat-uric** is due to a decrease in reabsorption of phosphate in the proximal tubules
- PTH also increases reabsorption of Ca^{2+} in the distal tubules of kidney

Calcium and phosphorus ...

- Ca^{2+} excretion is often increased in hyper-parathyroidism
 - because the amount Ca^{2+} filtered overwhelms the effect on reabsorption
- PTH also increases the formation of 1,25 dihydroxy Cholecalciferol, and this increases Ca^{2+} & phosphate absorption from intestine
- *On longer time scale*, PTH stimulates osteoclasts and osteoblasts with the effect osteoclasts predominating
 - so that more Ca^{2+} is mobilized from bone

Calcitonin:

- The normal plasma Ca^{+2} level is 9-10.5mg/dl
- Any disturbance of this can be result in diseases
- Calcitonin is not secreted until the plasma calcium level reaches approximately 10.5mg/dl and above this calcium level
- Plasma calcitonin is directly proportionate to plasma calcium
- Beta –adrenergic agonists, dopamine and estrogen, stimulate calcitonin secretion also
- Gastrin, CCK and secretin have all been reported to stimulate calcitonin secretion

Actions of Calcitonin:

- Serpentine receptors for calcitonin are
 - ✓ found in bones and the kidneys
- Calcitonin lowers the circulating calcium and phosphate levels.
- It exerts its calcium lowering effect by *inhibiting bone resorption*
- This action is direct, and calcitonin inhibits the activity of osteocalsts in vitro.

Calcium and phosphorus regulation...

- Circulating ionized calcium acts directly on the parathyroid gland in
 - a negative feedback fashion to regulate the secretion of PTH
- Increased plasma phosphate stimulates PTH secretion by lowering plasma Ca^{2+}
- Magnesium is required to maintain normal parathyroid secretory response
- Impaired PTH release, followed by impaired synt. Of 1, 25 dihydroxy cholecalciferol from target organ, accounts for the hypocalcemia
 - that occasionally occurs in magnesium deficiency

Role of vitamin D:

- The active transport of Ca^{2+} & PO_4 from intestine is increased by metabolite of vitamin D
- Vitamin D₃ which is also called cholecalciferol is produced in skin from 7-dehydrocholesterol by action of sunlight
- In liver vitamin D₃ is converted to 25 -hydroxycholecalciferol (25-OH D₃) /calcidiol .
- It is then converted to 1, 25 dihydroxycholecalciferol (calcitriol) in the proximal tubules of kidney

Actions of vitamin D:

- Calcitriol, in addition to increasing Ca^{2+} absorption from the intestine,
 - it also facilitates Ca^{2+} reabsorption in the kidneys
- It acts on bones, where it mobilize Ca^{2+} and secretes PO_4 , by increasing the number of mature Osteoclasts
- It also stimulates osteoblasts, but the net effect is still Ca^{2+} mobilization.

Actions of vitamin D:

Mechanism of actions:

- The formation of 1, 25 dihydrocholecalciferol in the kidneys which is catalyzed by 1 α hydroxylase is regulated in feed back fashion by plasma Ca^{2+} & PO_4 .
- the production of 1,25 dihydroxycholecalciferol with PTH is also increased by low Ca^{2+} level and
- inhibited by high plasma PO_4 levels,
 - by the direct inhibitory effect of PO_4 on alpha 1-hydroxylase

DISORDERS INVOLVING BONES

Rickets:

- It is due to deficiency of Vitamin D during childhood
- The full-blown condition in children is characterized by weakness and bowing of weight bearing bones, dental defects and hypocalcemia

Osteomalacia:

- It is due to deficiency of vitamin D during adulthood, results from
 - demineralization of bones, un able to absorb Ca^{+2} from intestine
- It is especially in women who have little exposure to sunlight, often several pregnancies.

Hyperparathyroidism:

- Excessive parathormone cause bone resorption

Osteogenesis imperfecta:

- Over 90% of patients with osteo-genesis imperfecta have mutation in genes
- It is brittle bone disease characterized by abnormal fragility of bones due to vit C deficiency
 - **decreased synthesis of the hydroxylated amino acids results**
- **Read on Osteoporesis**

Thank you!!